

Classroom **AV** Essentials

Day 2 — Troubleshooting and Continuity of Instruction

“When technology fails, your instruction doesn't have to.”



DAY 2 OF 2



7 HOURS



PROFESSIONAL DEVELOPMENT



ISTE x AVIXA

Day 2 at a Glance

7

Hours of Professional
Development

6

Core Modules + Capstone

1

Core Goal: Troubleshooting
Mastery

ISTE x AVIXA PD

Classroom AV Essentials
Day 2 of 2

TIME	SESSION	DURATION
8:00-8:30	OPENER Kit Review and Day 1 Recap	30 min
8:30-9:30	MODULE 7 The Troubleshooting Mindset	60 min
9:30-9:45	☺ Break	15 min
9:45-11:00	MODULE 8 No Image on Screen — Room PC & Laptop	75 min
11:00-12:00	MODULE 9 No Audio	60 min
12:00-1:00	🍽 Lunch	60 min
1:00-2:00	MODULE 10 Video Conferencing Failures	60 min
2:00-2:15	☺ Break	15 min
2:15-3:00	MODULE 11 Control System and Other Failures	45 min
3:00-3:45	MODULE 12 Escalation and IT Communication	45 min
3:45-4:30	CAPSTONE Scenario Stations	45 min

Day 1 Quick Recap

📖 WHAT WE COVERED YESTERDAY

- 📶 **The 4 AV Layers** — Source, Display, Audio, Control
- 📶 **Signal Flow Chain** — your troubleshooting map
- 📶 **Cable and Connector Types** — HDMI, USB-C, VGA, adapters
- 📶 **Your Proactive AV Tech Kit** — assembled and ready
- 📶 **Accessibility Standards** — AV as an equity requirement



📶 **Signal flow is your troubleshooting map** — we will use it all day.

MODULE 07

The Troubleshooting Mindset

A Systematic Approach That Saves Instructional Time

”

Random button-pressing wastes time, can worsen problems, and gives IT bad information.

🕒 60 MINUTES

• THE MOST IMPORTANT MODULE OF THE DAY

Four Questions to Ask First

1

WHAT exactly is wrong?

Black screen? No sound? Partial image? Error message?
Be **specific** — vague descriptions waste diagnostic time.



2

WHAT is connected?

Room PC or laptop? HDMI or USB-C? Adapter or direct?
Know your signal chain **before** you trace it.



3

WHEN did it start?

Just now? Yesterday? Only in this room? Only with this device?
Timing narrows the cause **dramatically**.



4

WHAT changed recently?

New cable? Software update? Room rearrangement? New device?
Changes cause failures — find the change, find the fix.



! Answering these four questions also gives IT **exactly what they need** — whether you fix it or not.

The **5-Step** Classroom Troubleshooting Process

01

STEP

**STOP AND OBSERVE**

What do you **see**? What do **LEDs indicate**? What is the **error message**? Gather information before acting.

02

STEP

**FOLLOW THE SIGNAL**

Source → Cable → Switcher → Display. Walk the chain from origin to output — identify exactly where it breaks.

03

STEP

**TRY SIMPLE THINGS FIRST**

Power cycle · Re-seat cable · Correct input · Win + P · Check batteries in remote. Solve the obvious before going deeper.

04

STEP

**TEST ONE VARIABLE AT A TIME**

Swap **ONE thing**. Changing three things simultaneously makes diagnosis **impossible** — you won't know what fixed it or what caused it.

05

STEP

**KNOW WHEN TO STOP**

Unresolved after **5 minutes**? Switch to a **workaround** AND submit an **IT ticket** simultaneously — then keep teaching. Instructional time cannot be recovered.



Speaker note: Walk through each step with a relatable classroom example. Use a live scenario: "Your laptop just connected and the projector shows nothing — walk me through your 5 steps." Cold-call a participant for each step.



THE 5-MINUTE RULE

- 1 Give yourself **5 minutes** on any AV problem.
- 2 If not fixed in 5 minutes — **STOP**. Switch to a workaround. **TEACH**.
- 3 Submit an IT ticket **at the next break**.

Instructional time is not recoverable. Your students' learning is always the priority.



IMMEDIATE ACTION REQUIRED — NO TROUBLESHOOTING

STOP — AND CALL FACILITIES / IT NOW

CALL
NOW

- ✗ 🔥 Burning smell or visible smoke from any AV equipment
- ✗ ⚡ Sparking or flickering from any power source
- ✗ 🔒 Open rack cabinet exposing live or energized equipment
- ✗ ⚙️ Equipment physically damaged or fallen — mount failure, broken housing
- ✗ 👤 Any person physically harmed by AV equipment — injury of any kind



For these situations — **do not troubleshoot**. Clear the area, protect students, then call. Your safety and your students' safety come first — always.

SPEAKER NOTE

2 min — Safety first

MODULE 08

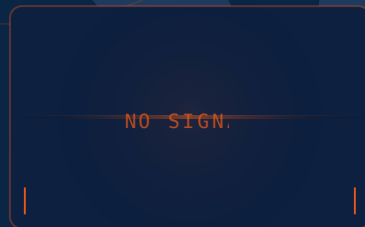
No Image on Screen

The Most Common Classroom AV Failure —
and the Most Solvable

THIS MODULE:

• Scenario A: Room PC

• Scenario B: Laptop





• No Image — Step by Step / 5 Steps to Restore Display Output

1

STEP 1**Check Display Power**

LED solid **green or blue**? ✓ On.

LED **amber, blinking, or off?**

→ Press power on control panel or display.

→ Wait **15 seconds**.

2

STEP 2**Select Correct Input**

Control panel or remote →

Input / Source

→ Select

'COMPUTER'

or **'HDMI 1'**

X NOT 'LAPTOP'

3

STEP 3**Toggle Output (Win+P)**

Press **Win + P**

→ Select **'Duplicate'**

→ Wait **5 seconds**

for display to re-detect signal.

4

STEP 4**Re-seat HDMI Cable**

Cable at **back of PC tower**.

Unplug → wait **3 sec**

→ reconnect **firmly**.

Check for **bent pins**.

5

STEP 5 — LAST RESORT**Reboot PC**

Restart PC with cable

still connected —

forces

display re-detection

on startup.

**KEEP TEACHING — DON'T WAIT**

Blank after Step 3? **Use your personal laptop on the LAPTOP input** while IT addresses the room PC — your students keep learning.

DEPLOY WORKAROUND

Submit IT ticket at next break

STEP
1**CHECK DISPLAY POWER****LED indicates display state**

Solid green or blue = on and ready.

Amber or blinking = press power on control panel or display. Wait 15 seconds.

STEP
2**SELECT CORRECT INPUT****Use LAPTOP or HDMI 2**

LAPTOP

HDMI 2

× NOT 'COMPUTER'

STEP
4**TOGGLE OUTPUT****Force display detection**

Win + P

STEP
5**TEST ADAPTER****Swap adapter from your kit**

Adapters **fail silently** — no error message, just no image. This is the **#1 cause** of laptop connection failures. Always carry a spare.

STEP
3**CONNECT CABLE****Seat firmly at BOTH ends**

Push HDMI or USB-C adapter fully into **laptop port** and **wall plate**. Listen for a click or firm stop. A half-seated cable produces no signal.

STEP
6**FIX RESOLUTION****Image shows but wrong size?**

1920 × 1080

**KEEP TEACHING**

Still no image after Step 5? **Switch to Room PC on COMPUTER input** — hand your students a LMS link to your slides and keep the lesson moving while IT investigates the laptop connection.



Speaker note: 10 min — Walk through each step with a laptop and adapter physically in hand. Demonstrate the Win+P shortcut and Mac Displays mirroring live.

SYMPTOM	MOST LIKELY CAUSE	QUICK FIX
 Black screen — display LED on	Wrong input selected on control panel	Select correct input on control panel →  
 "No Signal" message on display	Cable not connected or loose at one end	Re-seat cable at both ends → confirm correct input selected
 HDCP error or black flash on screen	Copy protection conflict between devices	Power cycle <i>all</i> devices in sequence: display → switcher → source
 Partial image or wrong size / cropped	Resolution mismatch between laptop and display	Right-click desktop → Display Settings → set to 
 Image on laptop only — not room display	Extended desktop mode active instead of Duplicate	 → select  → Displays → Mirror
 Image too dark or noticeably dim	Projector lamp near end-of-life or failing	Raise brightness via control panel → submit IT ticket for lamp replacement
 Signal dropout every few seconds	Failing or damaged HDMI cable	Swap cable immediately → use spare HDMI from your AV Tech Kit



Handbook reference: Every row in this table has a full step-by-step procedure on **p. 14-15** of your handbook. Keep it open at your AV station. • **5-Minute Rule:** If you cannot identify the fix in 5 minutes — deploy a workaround and submit a ticket.



No Audio

The Problem Most Teachers Try to Ignore —
And Shouldn't



! KEY INSIGHT

"Audio failure is an **equity and accessibility issue**, not just an inconvenience."



Audio Problem Quick Diagnosis

- Quick Fix — Try First
- Workaround + Report to IT
- Escalate / Submit Ticket

SYMPTOM	MOST LIKELY CAUSE	QUICK FIX
✓ No sound at all	Control panel muted or volume at zero	Press Mute button to toggle off, or raise volume on control panel
✓ No sound from laptop only	Audio routed to laptop's internal speakers	Taskbar speaker icon → click chevron → select HDMI Output or room display name
✓ No sound from video	In-app audio muted	Check browser tab mute + media player volume slider
✓ Microphone not working	Wrong input selected on DSP	Control panel → Mic Input selection; confirm correct mic source
⚠ Feedback / squealing	Mic too close to speakers	Move mic away or lower mic gain on control panel immediately
⚠ Low hum or buzz	Ground loop (laptop charger)	Disconnect laptop charger temporarily · Report to IT
❗ Echo on video calls	AEC (echo cancellation) not functioning	Use headset immediately · Report to IT — AEC requires service
❗ Audio from one side only	Speaker failure or cable fault	Submit ticket · Use portable Bluetooth speaker as workaround

Routing Laptop Audio to **Room Speakers**



WINDOWS

Restore Room Audio

1



Bottom-right system tray — look for the volume icon

2



Click the chevron › arrow next to the volume slider

This expands the output device selector list

3

**HDMI Output**

May appear as room display model name or "Digital Audio (HDMI)"

4



Test with a short audio clip

Play 5–10 seconds of any media — confirm sound from room speakers



MAC

Restore Room Audio

1

**Apple menu**

Click the Apple logo top-left on screen, then navigate to Sound

2



Click the Output tab

Switch from Input to Output to control speaker destination

3

**HDMI**

Room display appears as its model name or "LG TV HDMI" / "Epson HDMI"

4



Raise the Output Volume slider

Also ensure "Mute" checkbox is unchecked — test with audio playback

KEEP TEACHING



No room audio? A charged Bluetooth speaker costs \$30 and saves your lesson. **Your voice is always your most reliable audio source** — move to the center of the room and project.

Video Conferencing and Hybrid Failures

Remote Learners Deserve Equitable Access — Every Time

HYBRID EQUITY PRINCIPLE

"If a remote participant cannot **hear**, **see**, and **participate equally** — your hybrid classroom is not hybrid. It is two separate, **unequal** experiences."



In-Room



Remote



ACCESS IS EQUITY



60 Minutes



Classroom AV Essentials — Day 2



Troubleshooting and Continuity of Instruction



60 MINUTES

THIS MODULE COVERS

- Camera & Microphone Failures
- Platform Audio Routing Issues
- Bandwidth & Connectivity Drops
- Content Sharing & Caption Failures

ISTE x AVI

Before Every Call — **Your Pre-Flight Checklist**

⌚ 5 min

✓ SETUP & CONNECTION

- ✓ Join **5 minutes early** — always
Never start the call scrambling — you need buffer time
- ✓ Test **audio input and output** in platform settings
Speak → confirm mic shows activity; play tone → confirm room speakers respond
- ✓ Test **camera framing** — can remote participants see you and the board?
Check preview: show face, whiteboard, and student area if possible
- ✓ Test **content / screen sharing**
Share your slides now — not after the session starts

🛡️ EQUITY & FALLBACK

- ✓ Have **laptop fallback** open and logged in
If room PC fails mid-call, you switch in under 60 seconds
- ✓ Confirm **captioning is enabled**
Turn it on by default — not on request. It's an equity requirement
- ✓ Know **which camera shows the room** — not just you
Remote participants need to see student interaction, not only the instructor view

! Remote learners deserve the same access as in-room students — every session.



A 5-minute pre-flight check prevents 90% of video conferencing disasters. Run it silently while students settle — no disruption, maximum protection.

During-Call **Problem Solver**

❗ PROBLEM	🔍 MOST LIKELY CAUSE	✓ QUICK FIX
 Call drops immediately	Bandwidth or firewall restriction	 Switch to wired Ethernet
 Remote participants hear echo	Mic picking up room speakers	 Use headset immediately
 Can't hear remote participants	Wrong audio output selected	 Platform audio settings → select room speakers
 Camera shows black	Wrong camera selected in platform	 Platform settings → select correct camera
 Content share shows black or static	HDCP copy protection conflict	 Use non-HDCP source or share via app window
 Captions not appearing	Feature not enabled in platform	✓ Find CC button in platform toolbar → enable
 Video freezes repeatedly	Bandwidth insufficient	↩ Lower video quality or switch to audio-only

Before You Move On — Ask Yourself

1



HEAR

Can remote participants hear **EVERY** in-room speaker — including student voices?

☐

2



SEE

Can remote participants **SEE** the content area that matters right now?

☐

3



READ

Can remote participants **READ** all shared slides and board content?

☐

4



PARTICIPATE

Can remote participants **PARTICIPATE** equally — hand raise, chat, polling?

☐

5



CAPTION

Is captioning **ON by default** — not optional?

☐

If any answer is **No** — fix it before continuing. Hybrid equity is not optional.

Discussion: "Which of these fails most often in your classroom?"

Control System and Other Common Failures

When the Remote Does Nothing and the Panel Goes Dark



KEY INSIGHT

Control system failures **feel catastrophic** — most resolve in **60-90 seconds** with patience.

🕒 45 minutes

☑️ Touchpanel Failures

📱 Remote Not Responding

🖥️ Projector Warning Signs

Control System Quick Reference

✓ Try First

✗ Escalate

⌚ 8 MIN

! SYMPTOM	MOST LIKELY CAUSE	QUICK FIX	ESCALATE?
 Touchpanel black or off	Sleep mode / power off	Touch to wake. If no response after 10 sec, hold power button. Wait 30 sec.	 Escalate if 30 sec pass
 Panel on, buttons do nothing	Processor rebooting	Wait 60-90 seconds. Processor restarts automatically — do not power cycle.	 Escalate if 2 min pass
 Input selected; display didn't switch	Firmware or signal fault	Select input again. If no change, power cycle the display directly. Retry once.	 Escalate if repeated
 "System Unavailable" error	Control processor offline	Wait 2 minutes, then retry. Use manual display controls to	✗
 One function fails; rest works fine	Device lost network IP	Note which device/function failed. Continue using work	✗

⌚ **Key insight:** Control system failures feel catastrophic — most resolve in **60-90 seconds** with patience. Always try manual display controls as your workaround while the system recovers.

Projector-Specific Warning Signs

7 MIN

SYMPTOM	MEANING	WHAT TO DO	ACTION
 Lamp warning LED on	Lamp near end of life or overheating	Submit ticket immediately	
 Yellow or green color tint	Lamp degrading — color balance failing	Submit ticket Continue using until IT advises otherwise	
 Fan running unusually loud	Filter clogged or thermal overload	Submit ticket Check for ventilation obstructions	
 Projector shuts off mid-class	Thermal shutdown triggered	Clear ventilation · allow 15 min cool-down Then submit ticket	
 Black bars on all edges	Overscan or resolution mismatch	Set laptop resolution to 1920×1080	

**LAMP SAFETY WARNING**

Never attempt to replace a projector lamp yourself — high-pressure mercury lamps are hazardous.

Escalate**Always submit a ticket.**

Call or ticket IT now

Monitor

Submit ticket, continue using

Quick Fix

Resolve yourself

Reference: Handbook p. 34

12

Escalation & IT Communication

Knowing When to Stop — and How to Get Help Fast



KEY PRINCIPLE

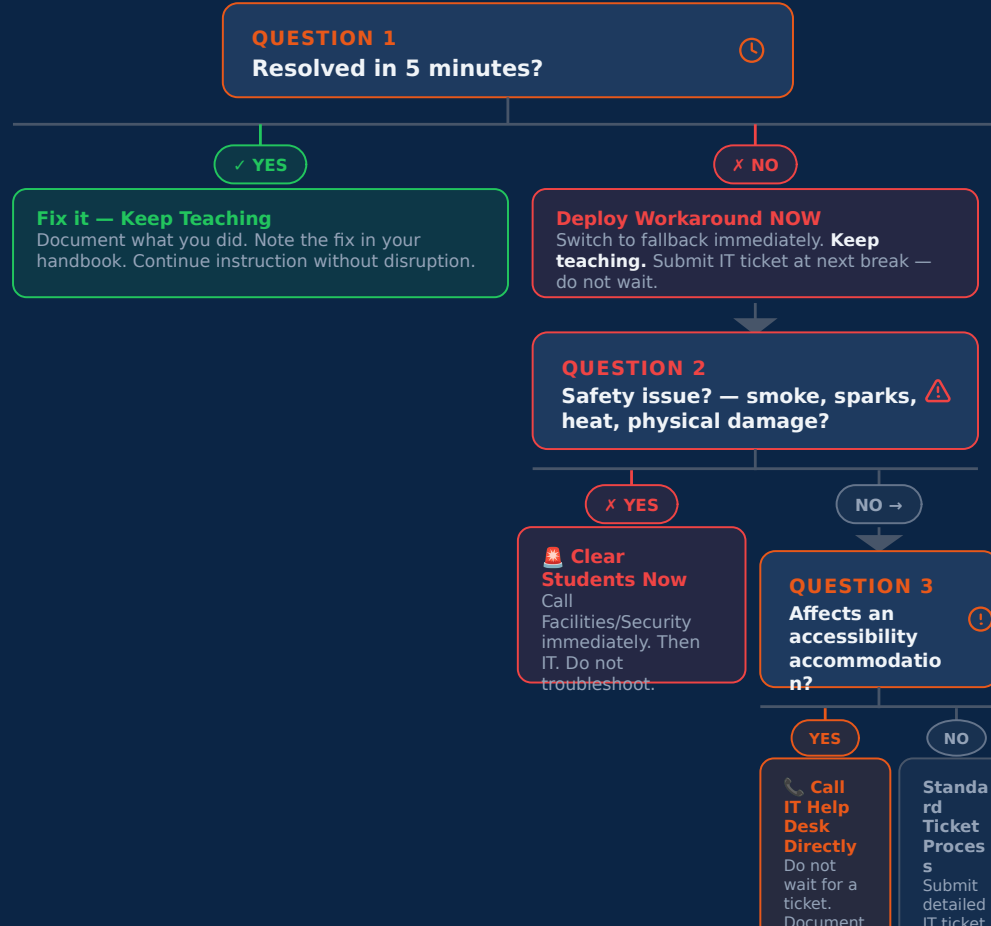
A **well-written IT ticket** is the fastest path to a permanent fix.

 45 minutes



🔔 The Escalation Decision Framework

Module 12 · When to keep going — and when to stop and get help



How to Write an AV IT Ticket **That Gets Fixed**

8 MIN

✗ ✗ BAD EXAMPLE — LOW PRIORITY TICKET

SUBJECT

Projector broken again.

BODY

The projector isn't working. Please fix.

MISSING INFORMATION

- ✗ Which room? Which device? Which connection?
- ✗ What was tried? What do LEDs show?
- ✗ Is this recurring? Accessibility affected?

⚠ Triaged as low priority — provides nothing useful to IT.

✓ ✓ GOOD EXAMPLE — FAST RESPONSE TICKET

SUBJECT

Room 204B — No image on projector when MacBook connected via HDMI, 6/11/24

BODY

Symptom: No image on projector.

Display: Power light solid green. Input set to LAPTOP/HDMI 2.

Tried: Win+P → Duplicate; re-seated cable; tried second HDMI cable from kit — still no image.

Connection: MacBook Air M2 → USB-C adapter → HDMI → wall plate.

Pattern: Consistent every session this week.

Accessibility: No accommodation currently affected.

Availability: 11am–2pm for access.

📍 Room ID 🖱️ Device + Connection 🔧 Steps Tried 🔄 Recurrence
♿ Accessibility 🕒 Availability

✓ Prioritized, diagnosed remotely, and fixed permanently.

📌 The details you include determine how fast IT responds — and whether the problem gets fixed for good.

When AV Fails, Your Lesson Doesn't Have To

WORKAROUND STRATEGIES — KEEP TEACHING

AV FAILURE	KEEP TEACHING WORKAROUND	TIME TO IMPLEMENT
 No projector image	Share slides via LMS on student devices	
 No room audio	Portable Bluetooth speaker ; narrate video yourself	
 Video conferencing failure	Phone audio bridge ; share slides via LMS link	
 Room PC failure	Personal laptop on LAPTOP input	
 Control system unresponsive	Manual controls directly on display	
 No microphone	Project voice; move to front and center of room	
 Document camera offline	Phone camera on stand via Bluetooth or USB	
 Total AV failure	Whiteboard + handouts ; pivot to Socratic discussion	



The 5-Minute Rule applies here too. If an AV fix takes more than 5 minutes — stop, deploy your workaround, and keep your students learning. Every workaround above can be ready **in 3 minutes or less**. Instructional time is not recoverable.

CAPSTONE — SCENARIO STATIONS

🕒 45 Minutes | 4 Stations | 10 Minutes Each

Rotate through all four stations. Each station has a scenario card. Use your handbook and kit to **diagnose and resolve** — or **decide to escalate**.



STATION 01

The Blank Screen

Room PC + projector display · Signal path diagnosis

ORANGE BADGE



STATION 02

The Silent Video

Laptop audio routing · Output device selection

BLUE BADGE



STATION 03

The Frozen Call

Video conferencing failure · Bandwidth + quality

TEAL BADGE



STATION 04

The Unresponsive Panel

Control system failure · Wait vs. escalate decision

PURPLE BADGE

📄 COMPLETE DIAGNOSTIC FORM IN YOUR HANDBOOK FOR EACH STATION

Symptom

First thing checked

Root cause

Fix or workaround

Escalate?

IT ticket wording



Brief setup instructions (~3 min) — explain rotation process, hand out scenario cards, confirm handbook page. Then say "Go!" and let participants rotate every 10 minutes

STATION 1 🖥️ The Blank Screen

SCENARIO

Room PC is connected. Display power light is **solid green**. Control panel shows **HDMI 2** selected. No image appears on projector.

✓ EXPECTED FIX

Wrong input selected — switch to **COMPUTER / HDMI 1** on control panel.

STATION 2 🔊 The Silent Video

SCENARIO

Laptop is connected and image is working perfectly. A YouTube video is playing but **no audio** comes from room speakers — only from the laptop.

✓ EXPECTED FIX

Audio routed to laptop speakers — taskbar speaker icon → **select HDMI Output** for room audio.

STATION 3 📞 The Frozen Call

SCENARIO

Video conference call drops video **every 20 seconds**. Remote participants hear audio but see only a frozen frame. Audio is consistent.

✓ EXPECTED FIX

Bandwidth issue — **switch to wired Ethernet** and lower video quality in platform settings.

STATION 4 📱 The Unresponsive Panel

SCENARIO

Touchpanel is **lit and powered on**. Pressing any button does nothing. Inputs need to be changed but the panel is unresponsive.

✓ EXPECTED FIX

Control processor rebooting — **wait 60-90 sec**, then try manual controls directly on the display.



For each station, complete the diagnostic form in your handbook:

SYMPTOM

FIRST CHECK

ROOT CAUSE

FIX / WORKAROUND

ESCALATE?

IT TICKET WORDING

You Did It — Two Days Well Spent

DAY 2 COMPLETE

ISTE x AVIXA PD Initiative



DAY 1

What You Learned

- **4 layers** of every AV system
- Signal flow & troubleshooting map
- Cable types & the **Educator's AV Kit**
- Accessibility as an **AV requirement**

🕒 7 Hours • Modules 1-6



DAY 2

What You Can Now Do

- Diagnose & fix **No Image** in under 5 min
- Restore lost audio **step by step**
- Handle **video conferencing** failures systematically
- Write **IT tickets** that get fast responses
- Deploy **workarounds** that keep instruction moving

🕒 7 Hours • Modules 7-12 + Capstone



WHAT HAPPENS NEXT

Your action checklist

- ✓ Keep your **handbook** at your AV station
- ✓ Complete your **AV Tech Kit**
- ✓ Add **IT Help Desk** number to your phone
- ✓ **Laminate** your Quick Reference Cards
- ✓ Submit a **diagnostic IT ticket** for any chronic issue in your room

✓ You are ready. Start Monday.

🗨 Speaker note: 10 min — Q&A, congratulate participants, distribute certificates if applicable. Pause for reflection.



"Technology should serve learning. You are now equipped to make sure it does."

Your Continuing AV Resources

Everything you need — long after class ends.

THANK YOU
Day 2
Complete



This Handbook

Every table, checklist, and step-by-step guide in this course is in your handbook — your **permanent classroom reference**.

✓ **Keep at your AV station**



Quick Reference Cards

Laminated **No Image** and **No Audio** cards — keep at your AV station or inside your Educator's AV Kit.

✓ **Laminate for longevity**



ISTE & AVIXA Standards

Full standards and continuing professional development resources for educators.

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